# Gengcong Yang

gengcong.y@gmail.com | GitHub | LinkedIn | Google Scholar

## Education

**University of Texas at Austin (UT Austin)** 08/2023 – 12/2024
*Master in **Computer Science**; GPA: 3.7 / 4.0* Austin, United States

**Tsinghua University (THU)** 08/2019 – 06/2022
*Master in **Control Engineering**, Department of Automation; GPA: 3.6 / 4.0* Beijing, China
*Advisor: Yujiu Yang*

**National Chiao Tung University (NCTU)** 09/2017 – 01/2018
*Exchange Student in **Computer Science**; GPA: 3.8 / 4.0* Hsinchu, Taiwan

**Sun Yat-sen University (SYSU)** 08/2015 – 06/2019
*Bachelor in **Software Engineering**; GPA: 3.8 / 4.0* Guangzhou, China

## Publications

**Diamond: Harnessing GPU Resources for Scientific Deep Learning.** *IEEE eScience*, 2025.

Haotian Xie, Rohan Marwaha, Minu Mathew, Song Bian, **Gengcong Yang**, et al.

- Contributed to the design of a distributed job-submission pipeline enabling scalable, multi-node GPU training on a platform for the scientific machine learning lifecycle; funded by **NSF**.

**Probabilistic Modeling of Semantic Ambiguity for Scene Graph Generation.** *CVPR*, 2021.

**Gengcong Yang**, Jingyi Zhang, Yong Zhang, Baoyuan Wu, Yujiu Yang.

- Proposed a probabilistic framework and designed a plug-and-play uncertainty modeling module to capture semantic ambiguity and multiple plausible visual relationships in scene graphs.

## Research Experience

**Texas Advanced Computing Center**, *Graduate Research Assistant* Austin, 01/2024 – 12/2024

- Supervisor: Yinzhi Wang
- Contributed to the early-stage development of Diamond, an NSF-funded, scalable platform supporting the full machine learning research lifecycle on HPC resources.
- Designed and implemented a distributed job-submission pipeline enabling users to containerize environments and train ML models across multiple GPU nodes.

**KAUST**, *Research Intern* Remote, 02/2021 – 07/2021

- Supervisor: Bernard Ghanem
- Implemented a differentiable RoI Pooling method with CUDA and C++ for the temporal action detection task on videos, achieving an experimental improvement of 6%.

**Tencent AI Lab**, *Computer Vision Researcher Intern* Shenzhen, 07/2019 – 07/2020

- Supervisor: Baoyuan Wu
- Developed a novel modeling method for polysemous visual-relationship representation and applied for a technology patent.
- Contributed to the design and experimentation of the probabilistic uncertainty modeling framework later published at CVPR 2021.

## Industry Experience

**TikTok USDS**, Emerging Product, *Machine Learning Engineer* San Jose, 07/2026 – Present

- Led safety guardrails to detect and block unsafe prompts and generated videos for Seedance, a video generation model.
- Built and scaled GPU model-serving infrastructure for U.S. deployment.
- Developed LLM-based agents that automate the team's operational workflows, reducing manual effort.

**Uber**, Driver Offer Pricing Team, *Machine Learning Engineer* Sunnyvale, 01/2025 – 07/2026

- Developed driver fare prediction models used as core pricing signals within Uber's marketplace pricing system.
- Improved observability and reliability infrastructure for pricing prediction pipelines running at scale.

**ByteDance**, E-commerce, *Machine Learning Engineer* Shanghai, 06/2022 – 06/2023

- Developed image search models and a retrieval-ranking framework for the TikTok Shop similar-product search project.
- Constructed large-scale datasets for post-pretraining and quantized object detection models to improve serving efficiency.

**Tencent**, *Machine Learning Engineering Intern* Beijing, 06/2021 – 08/2021

- Focused on the retrieval stage of the recommendation system for an early-stage short-video platform.
- Built an offline evaluation pipeline for retrieval models and experimented with pairwise training for a two-tower model.

## Patents

**Visual Relationship Recognition Method Based on Artificial Intelligence**

- Inventors: **Gengcong Yang**, Yong Zhang, Baoyuan Wu, Yanbo Fan, Zhifeng Li, Yujiu Yang, Wei Liu
- Patent Number: CN202011108484
- Granted Date: 03/05/2024

## Selected Awards

| | |
|---|---|
| Tsinghua SIGS Professional Practice Scholarship | 2022 |
| Tsinghua Excellent Scholarship First Prize | 2021 |
| SYSU Outstanding Undergraduate Thesis Award | 2019 |
| SYSU Excellent Scholarship Second Prize | 2018 |
| National Encouragement Scholarship | 2017, 2018 |
| SYSU Xian Weijian Overseas Study Funding | 2017 |
| Interdisciplinary Contest In Modeling Honorable Mention | 2017 |
| SYSU Excellent Scholarship Third Prize | 2016, 2017 |